KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 13 · MATH

Market Day in the Andes — Teacher Lesson Plan & Answer Key

Learning objectives: Students multiply a whole number by a fraction, multiply a fraction by a fraction using an area model and the numerator-times-numerator, denominator-times-denominator rule, simplify products with the greatest common factor, and solve one- and two-step word problems with mixed numbers. Students also judge whether a product is reasonable, find mistakes in others' work, and correct capitalization, punctuation, and grammar in math sentences.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Killa in her Quechua village in the Andes of Peru, above the Sacred Valley near Pisac; student types first and last name so results are sent.
2. Slide 1 — Warm-Up: Week 12 Review (12 pts): equivalent fractions, simplifying, and mixed numbers / improper fractions. If a student misses two or more, rework one on paper before moving on; today's lesson depends on these skills.
3. Slide 2 — Pencil Break (Pre): student completes the Pre-Lesson Warm-Up sheet off screen.
4. Slide 3 — Hook: farming in the Andes (Inca terraces called andenes, thousands of native potatoes, quinoa, alpaca wool and natural dyes, market day); Killa's to-do list previews every problem type.
5. Slide 4 — Picture It: 4 × 2/3 cups of quinoa shown as 4 bars of thirds, then regrouped into 2 2/3. Stress that the denominator stays the same because the size of the pieces does not change; check with repeated addition.
6. Slide 5 — Practice: a whole number × a fraction (9 pts).
7. Slide 6 — Teach: the area model for 3/4 × 2/5 on a potato terrace (6 of 20 parts = 6/20 = 3/10), then the shortcut: multiply numerators, multiply denominators. Ask: why is the product smaller than both fractions?
8. Slide 7 — Practice: a fraction × a fraction, including reading an area model (9 pts).
9. Slide 8 — Teach: simplifying products with the GCF (3/8 × 4/9 = 12/72 = 1/6; 4 × 5/6 = 20/6 = 3 1/3), the simplify-first shortcut, and the trap of dividing by a common factor that is not the greatest.
10. Slide 9 — Practice: simplify the product, including a find-the-mistake item (9 pts).
11. Slide 10 — Vocabulary flip cards: numerator, denominator, product, area model, simplest form, improper fraction, mixed number, GCF.
12. Slide 11 — Vocabulary check (6 pts).
13. Slide 12 — Teach: mixed numbers at the Pisac market. 3 × 2 1/2 two ways (improper fraction, or break apart), and 2/3 × 2 1/4 = 1 1/2. Warn against multiplying only the whole-number part.
14. Slide 13 — Practice: mixed numbers (9 pts).
15. Slide 14 — Real World: Grandmother Rosa's alpaca-yarn weaving, two steps (multiply, then subtract from 2 kilograms) (5 pts).
16. Slide 15 — Real World: the quinoa harvest on a terrace, two steps (fraction of a fraction, then subtract) (5 pts). Encourage a quick area-model sketch.
17. Slide 16 — Think It Through: "which explanation is correct?" and "find the mistake" (10 pts). Ask the student to say why each wrong choice is wrong.

18. Slide 17 — Fix the Sentence (8 pts): four sentences on quotation marks in dialogue, capitalizing proper nouns (places, languages, days), its vs. it's, commas in a series, and subject-verb agreement. The answer must match exactly (straight or curly quotes both work).
19. Slide 18 — Match-Up Round (10 pts): 5 clue-to-answer matches in simplest form. First-try matches earn 2 points each, 10 points in all.
20. Slide 19 — Speed Trail (bonus XP only).
21. Slide 20 — Looking Back reflection (bonus XP only).
22. Slide 21 — Boss Battle (8 pts): Sunday market in Pisac, a whole number × a mixed number, then a fraction of the total.
23. Slide 22 — Pencil Break (Post): student completes the Post-Lesson Challenge sheet off screen.
24. Slide 23 — Results: score out of 100 sent to KingCretot Experience; 85+ shows the market-day congratulations with a big confetti burst; below 85 shows encouragement, and Play Again gives the next question set.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: WEEK 12 REVIEW (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Equivalent fractions (Week 12): which fraction is equivalent to 2/3?	**4/6**	Multiply the numerator and the denominator by the same number: 2 × 2 = 4 and 3 × 2 = 6, so 2/3 = 4/6.
r2_s0	Set 1: Simplifying (Week 12): type 8/20 in simplest form. 8/20 =	**2/5**	The greatest common factor of 8 and 20 is 4. 8 ÷ 4 = 2 and 20 ÷ 4 = 5, so 8/20 = 2/5.
r3_s0	Set 1: Mixed numbers (Week 12): write 9/5 as a mixed number.	**1 4/5**	9 ÷ 5 = 1 with 4 left over (a remainder). That is 1 whole and 4 fifths, so 9/5 = 1 4/5.
r4_s0	Set 1: Equivalent fractions (Week 12): type the missing numerator. 3/5 = ?/20 ? =	**12**	5 × 4 = 20, so multiply the numerator by 4 too: 3 × 4 = 12. 3/5 = 12/20.
r1_s1	Set 2: Equivalent fractions (Week 12): which fraction is equivalent to 5/6?	**15/18**	Multiply the numerator and the denominator by 3: 5 × 3 = 15 and 6 × 3 = 18, so 5/6 = 15/18.
r2_s1	Set 2: Simplifying (Week 12): type 21/28 in simplest form. 21/28 =	**3/4**	The greatest common factor of 21 and 28 is 7. 21 ÷ 7 = 3 and 28 ÷ 7 = 4, so 21/28 = 3/4.
r3_s1	Set 2: Mixed numbers (Week 12): write 19/4 as a mixed number.	**4 3/4**	19 ÷ 4 = 4 with a remainder of 3. That is 4 wholes and 3 fourths left over, so 19/4 = 4 3/4.
r4_s1	Set 2: Equivalent fractions (Week 12): type the missing numerator. 2/3 = ?/12 ? =	**8**	3 × 4 = 12, so multiply the numerator by 4 too: 2 × 4 = 8. 2/3 = 8/12.
r1_s2	Set 3: Equivalent fractions (Week 12): which fraction is equivalent to 1/4?	**3/12**	Multiply the numerator and the denominator by 3: 1 × 3 = 3 and 4 × 3 = 12, so 1/4 = 3/12.
r2_s2	Set 3: Simplifying (Week 12): type 16/20 in simplest form. 16/20 =	**4/5**	The greatest common factor of 16 and 20 is 4. 16 ÷ 4 = 4 and 20 ÷ 4 = 5, so 16/20 = 4/5.

ID	Question	Answer	Why (explain to student)
r3_s2	Set 3: Mixed numbers (Week 12): write 2 1/5 as an improper fraction.	**11/5**	2 wholes = 10 fifths. Add the 1 fifth: 10/5 + 1/5 = 11/5.
r4_s2	Set 3: Equivalent fractions (Week 12): type the missing numerator. 4/5 = ?/15 ? =	**12**	5 × 3 = 15, so multiply the numerator by 3 too: 4 × 3 = 12. 4/5 = 12/15.

YOUR TURN: A WHOLE NUMBER × A FRACTION (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
a1_s0	Set 1: Type the product. (Either form of the answer is fine.) 3 × 2/9 =	**6/9 (also accept 2/3)**	Multiply the whole number by the numerator and keep the denominator: 3 × 2 = 6, so 3 × 2/9 = 6/9. That simplifies to 2/3.
a2_s0	Set 1: 4 × 3/5 = ?	**12/5**	4 groups of 3 fifths is 12 fifths: 4 × 3 = 12, and the denominator stays 5. 12/20 is wrong because the whole number multiplies only the numerator.
a3_s0	Set 1: Which mixed number is equal to 5 × 3/4?	**3 3/4**	5 × 3 = 15, so 5 × 3/4 = 15/4. 15 ÷ 4 = 3 with a remainder of 3, so 15/4 = 3 3/4.
a1_s1	Set 2: Type the product. (Either form of the answer is fine.) 4 × 2/10 =	**8/10 (also accept 4/5)**	Multiply the whole number by the numerator and keep the denominator: 4 × 2 = 8, so 4 × 2/10 = 8/10. That simplifies to 4/5.
a2_s1	Set 2: 3 × 4/7 = ?	**12/7**	3 groups, each with 4 sevenths, make 12 sevenths. 3 × 4 = 12, and the denominator stays 7: 3 × 4 = 12, and the denominator stays 7. 12/21 is wrong because the whole number multiplies only the numerator.
a3_s1	Set 2: Which mixed number is equal to 6 × 2/5?	**2 2/5**	6 × 2 = 12, so 6 × 2/5 = 12/5. 12 ÷ 5 = 2 with a remainder of 2, so 12/5 = 2 2/5.
a1_s2	Set 3: Type the product. (Either form of the answer is fine.) 2 × 3/8 =	**6/8 (also accept 3/4)**	Multiply the whole number by the numerator and keep the denominator: 2 × 3 = 6, so 2 × 3/8 = 6/8. That simplifies to 3/4.
a2_s2	Set 3: 5 × 2/3 = ?	**10/3**	5 groups of 2 thirds is 10 thirds: 5 × 2 = 10, and the denominator stays 3. 10/15 is wrong because the whole number multiplies only the numerator.
a3_s2	Set 3: Which mixed number is equal to 7 × 2/3?	**4 2/3**	7 × 2 = 14, so 7 × 2/3 = 14/3. 14 ÷ 3 = 4 with a remainder of 2, so 14/3 = 4 2/3.

YOUR TURN: A FRACTION × A FRACTION (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: Type the product. 1/2 × 3/5 =	**3/10**	Multiply the numerators: 1 × 3 = 3. Multiply the denominators: 2 × 5 = 10. The product is 3/10.

ID	Question	Answer	Why (explain to student)
b2_s0	Set 1: The area model shows 2/3 × 3/4. The columns show 3/4 (blue) and the rows show 2/3 (red). What fraction of the whole rectangle is shaded BOTH colors (gold)? [area model picture]	**6/12**	The rectangle has 4 × 3 = 12 equal parts. 3 columns times 2 rows = 6 parts are gold, so the product is 6/12. (You can simplify it to 1/2.)
b3_s0	Set 1: Type the product. 2/3 × 4/5 =	**8/15**	Multiply the numerators: 2 × 4 = 8. Multiply the denominators: 3 × 5 = 15. The product is 8/15.
b1_s1	Set 2: Type the product. 1/3 × 2/5 =	**2/15**	Multiply the numerators: 1 × 2 = 2. Multiply the denominators: 3 × 5 = 15. The product is 2/15.
b2_s1	Set 2: The area model shows 1/2 × 3/4. The columns show 3/4 (blue) and the rows show 1/2 (red). What fraction of the whole rectangle is shaded BOTH colors (gold)? [area model picture]	**3/8**	The rectangle has 4 × 2 = 8 equal parts. 3 columns times 1 row = 3 parts are gold, so the product is 3/8.
b3_s1	Set 2: Type the product. 3/4 × 3/5 =	**9/20**	Multiply the numerators: 3 × 3 = 9. Multiply the denominators: 4 × 5 = 20. The product is 9/20.
b1_s2	Set 3: Type the product. 1/4 × 3/5 =	**3/20**	Multiply the numerators: 1 × 3 = 3. Multiply the denominators: 4 × 5 = 20. The product is 3/20.
b2_s2	Set 3: The area model shows 2/3 × 2/5. The columns show 2/5 (blue) and the rows show 2/3 (red). What fraction of the whole rectangle is shaded BOTH colors (gold)? [area model picture]	**4/15**	The rectangle has 5 × 3 = 15 equal parts. 2 columns times 2 rows = 4 parts are gold, so the product is 4/15.
b3_s2	Set 3: Type the product. 5/6 × 1/2 =	**5/12**	Multiply the numerators: 5 × 1 = 5. Multiply the denominators: 6 × 2 = 12. The product is 5/12.

YOUR TURN: SIMPLIFY THE PRODUCT (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Multiply, then type the product in simplest form. 2/3 × 3/8 =	**1/4**	2 × 3 = 6 and 3 × 8 = 24, so the product is 6/24. The GCF of 6 and 24 is 6: 6 ÷ 6 = 1 and 24 ÷ 6 = 4. Simplest form: 1/4.
c2_s0	Set 1: 6 × 3/4 = ? Choose the answer in simplest form.	**4 1/2**	6 × 3 = 18, so the product is 18/4. 18 ÷ 4 = 4 with a remainder of 2, which is 4 2/4. 2/4 simplifies to 1/2, so the simplest form is 4 1/2.
c3_s0	Set 1: Find the mistake: Inti says 4/5 × 5/6 = 20/30 and stops there. What should he do?	**Simplify: the GCF of 20 and 30 is 10, so 20/30 = 2/3**	20/30 is correct but not finished. The GCF of 20 and 30 is 10, so divide both by 10 to get 2/3. Dividing by 5 gives 4/6, which can still be simplified.
c1_s1	Set 2: Multiply, then type the product in simplest form. 3/4 × 2/9 =	**1/6**	3 × 2 = 6 and 4 × 9 = 36, so the product is 6/36. The GCF of 6 and 36 is 6: 6 ÷ 6 = 1 and 36 ÷ 6 = 6. Simplest form: 1/6.

ID	Question	Answer	Why (explain to student)
c2_s1	Set 2: 8 × 5/6 = ? Choose the answer in simplest form.	**6 2/3**	8 × 5 = 40, so the product is 40/6. 40 ÷ 6 = 6 with 4 left over (a remainder), which is 6 4/6. 4/6 simplifies to 2/3, so the simplest form is 6 2/3.
c3_s1	Set 2: Find the mistake: Rosa says 3/10 × 5/6 = 15/60, and then she simplifies it to 3/12. What did she do wrong?	**3/12 can still be simplified. The GCF of 15 and 60 is 15, so 15/60 = 1/4**	Rosa divided by 5, but 5 is not the GREATEST common factor. The GCF of 15 and 60 is 15: 15 ÷ 15 = 1 and 60 ÷ 15 = 4, so the simplest form is 1/4. (3/12 also simplifies to 1/4.)
c1_s2	Set 3: Multiply, then type the product in simplest form. 5/8 × 2/5 =	**1/4**	5 × 2 = 10 and 8 × 5 = 40, so the product is 10/40. The GCF of 10 and 40 is 10: 10 ÷ 10 = 1 and 40 ÷ 10 = 4. Simplest form: 1/4.
c2_s2	Set 3: 9 × 2/3 = ? Choose the answer in simplest form.	**6**	9 × 2 = 18, so the product is 18/3. 18 ÷ 3 = 6 with no remainder, so the product is the whole number 6.
c3_s2	Set 3: Find the mistake: Mateo says 2/3 × 9/10 = 18/30, and then he simplifies it to 9/15. What did he do wrong?	**9/15 can still be simplified. The GCF of 18 and 30 is 6, so 18/30 = 3/5**	Mateo divided by 2, but 2 is not the GREATEST common factor. The GCF of 18 and 30 is 6: 18 ÷ 6 = 3 and 30 ÷ 6 = 5, so the simplest form is 3/5. (9/15 also simplifies to 3/5.)

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: A fraction whose numerator and denominator have no common factor except 1 is in ___.	**Simplest form**	When the only common factor is 1, the fraction cannot be simplified any more, so it is in simplest form.
v2_s0	Set 1: Type your answer: in the fraction 2/5, the 5 is the ___.	**denominator**	The bottom number tells how many equal parts make one whole. It is the denominator.
v1_s1	Set 2: A fraction like 7/4, with a numerator greater than its denominator, is called ___.	**An improper fraction**	When the numerator is greater than (or equal to) the denominator, the fraction is an improper fraction. 7/4 = 1 3/4.
v2_s1	Set 2: Type your answer: the answer to a multiplication problem is called the ___.	**product**	In 3 × 2/9 = 6/9, the answer 6/9 is the product.
v1_s2	Set 3: A rectangle split into rows and columns to show multiplying two fractions is called ___.	**An area model**	An area model splits a rectangle into rows and columns. The overlap shows the product.
v2_s2	Set 3: Type your answer: in the fraction 3/8, the 3 is the ___.	**numerator**	The top number tells how many equal parts we have. It is the numerator.

YOUR TURN: MIXED NUMBERS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
d1_s0	Set 1: Step one for mixed numbers: type 2 1/3 as an improper fraction. 2 1/3 =	**7/3**	2 wholes = 6 thirds. Add the 1 third: 6/3 + 1/3 = 7/3. (Shortcut: 2 × 3 + 1 = 7.)

ID	Question	Answer	Why (explain to student)
d2_s0	Set 1: Killa packs 4 bags of quinoa. Each bag holds 1 1/2 kilograms. How many kilograms is that in all?	**6 kilograms**	1 1/2 = 3/2, and 4 × 3/2 = 12/2 = 6. Or break it apart: 4 × 1 = 4 and 4 × 1/2 = 2, so 4 + 2 = 6 kilograms.
d3_s0	Set 1: 1/2 × 2 1/2 = ?	**1 1/4**	Change 2 1/2 to 5/2. Then 1/2 × 5/2 = 5/4, and 5/4 = 1 1/4.
d1_s1	Set 2: Step one for mixed numbers: type 3 1/4 as an improper fraction. 3 1/4 =	**13/4**	3 wholes = 12 fourths. Add the 1 fourth: 12/4 + 1/4 = 13/4. (Shortcut: 3 × 4 + 1 = 13.)
d2_s1	Set 2: Killa packs 3 bags of quinoa. Each bag holds 2 1/4 kilograms. How many kilograms is that in all?	**6 3/4 kilograms**	2 1/4 = 9/4, and 3 × 9/4 = 27/4 = 6 3/4. Or break it apart: 3 × 2 = 6 and 3 × 1/4 = 3/4, so 6 + 3/4 = 6 3/4 kilograms.
d3_s1	Set 2: 1/3 × 1 1/2 = ?	**1/2**	Change 1 1/2 to 3/2. Then 1/3 × 3/2 = 3/6, and 3/6 = 1/2.
d1_s2	Set 3: Step one for mixed numbers: type 4 2/5 as an improper fraction. 4 2/5 =	**22/5**	4 wholes = 20 fifths. Add the 2 fifths: 20/5 + 2/5 = 22/5. (Shortcut: 4 × 5 + 2 = 22.)
d2_s2	Set 3: Killa packs 5 bags of quinoa. Each bag holds 1 3/4 kilograms. How many kilograms is that in all?	**8 3/4 kilograms**	1 3/4 = 7/4, and 5 × 7/4 = 35/4 = 8 3/4. Or break it apart: 5 × 1 = 5 and 5 × 3/4 = 15/4 = 3 3/4, so 5 + 3 3/4 = 8 3/4 kilograms.
d3_s2	Set 3: 3/4 × 2 2/3 = ?	**2**	Change 2 2/3 to 8/3. Then 3/4 × 8/3 = 24/12, and 24/12 = 2.

REAL WORLD: GRANDMOTHER ROSA'S WEAVING (5 POINTS)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Killa's grandmother, Rosa, weaves shawls from alpaca yarn. Each shawl uses 3/8 kilogram of yarn. Rosa starts with 2 kilograms of yarn and weaves 4 shawls. Two steps: find how much yarn the shawls use, then type how many kilograms of yarn are LEFT. Kilograms left:	**1/2 kilograms (also accept 2/4, 4/8, 0.5, with or without the unit kg / kilogram(s))**	Step 1: 4 × 3/8 = 12/8 = 1 4/8 = 1 1/2 kilograms used. Step 2: 2 − 1 1/2 = 1/2 kilogram left.
w1_s1	Set 2: Killa's grandmother, Rosa, weaves shawls from alpaca yarn. Each shawl uses 2/5 kilogram of yarn. Rosa starts with 2 kilograms of yarn and weaves 3 shawls. Two steps: find how much yarn the shawls use, then type how many kilograms of yarn are LEFT. Kilograms left:	**4/5 kilograms (also accept 8/10, 0.8, with or without the unit kg / kilogram(s))**	Step 1: 3 × 2/5 = 6/5 = 1 1/5 kilograms used. Step 2: 2 − 1 1/5 = 4/5 kilogram left.
w1_s2	Set 3: Killa's grandmother, Rosa, weaves scarves from alpaca yarn. Each scarf uses 1/6 kilogram of yarn. Rosa starts with 2 kilograms of yarn and weaves 8 scarves. Two steps: find how much yarn the scarves use, then type how many kilograms of yarn are LEFT. Kilograms left:	**2/3 kilograms (also accept 4/6, with or without the unit kg / kilogram(s))**	Step 1: 8 × 1/6 = 8/6 = 1 2/6 = 1 1/3 kilograms used. Step 2: 2 − 1 1/3 = 2/3 kilogram left.

REAL WORLD: THE QUINOA HARVEST (5 POINTS)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: On one terrace, 3/4 of the field is planted with quinoa. On Monday, Killa's family harvests 2/3 of the quinoa part of the field. Two steps: what fraction of the WHOLE field is quinoa that has NOT been harvested yet?	**1/4**	Step 1: harvested = 2/3 × 3/4 = 6/12 = 1/2 of the whole field. Step 2: quinoa not harvested = 3/4 − 1/2 = 3/4 − 2/4 = 1/4 of the field.
w2_s1	Set 2: On one terrace, 2/3 of the field is planted with quinoa. On Monday, Killa's family harvests 1/4 of the quinoa part of the field. Two steps: what fraction of the WHOLE field is quinoa that has NOT been harvested yet?	**1/2**	Step 1: harvested = 1/4 × 2/3 = 2/12 = 1/6 of the whole field. Step 2: quinoa not harvested = 2/3 − 1/6 = 4/6 − 1/6 = 3/6 = 1/2 of the field.
w2_s2	Set 3: On one terrace, 4/5 of the field is planted with quinoa. On Monday, Killa's family harvests 3/4 of the quinoa part of the field. Two steps: what fraction of the WHOLE field is quinoa that has NOT been harvested yet?	**1/5**	Step 1: harvested = 3/4 × 4/5 = 12/20 = 3/5 of the whole field. Step 2: quinoa not harvested = 4/5 − 3/5 = 1/5 of the field.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? Killa multiplies 6 × 2/3 and gets 4. Is her answer reasonable?	**Yes. 2/3 is less than 1, so 6 groups of 2/3 is less than 6. 6 × 2/3 = 12/3 = 4.**	6 × 2/3 = 12/3 = 4, and it makes sense: each group is less than 1 whole, so 6 groups must be less than 6. Multiplying by a fraction less than 1 gives a smaller number.
x2_s0	Set 1: Find the mistake: Inti says 2/5 × 3/5 = 6/5. What did he do wrong?	**He did not multiply the denominators. 2/5 × 3/5 = 6/25.**	When you multiply fractions, multiply the numerators (2 × 3 = 6) AND the denominators (5 × 5 = 25). Keeping the same denominator is a rule for adding, not multiplying.
x1_s1	Set 2: Which explanation is correct? Is 3/4 × 5/6 greater than or less than 5/6?	**Less, because you are taking 3/4 of 5/6, which is only part of it**	3/4 × 5/6 means 3/4 of 5/6. Taking part of something gives less than the whole thing. Check: 3/4 × 5/6 = 15/24 = 5/8, and 5/8 is less than 5/6.
x2_s1	Set 2: Find the mistake: Rosa says 2 × 3 1/4 = 6 1/4. What did she do wrong?	**She did not multiply the 1/4. 2 × 3 1/4 = 6 + 2/4 = 6 1/2.**	Both parts of the mixed number get multiplied: 2 × 3 = 6 and 2 × 1/4 = 2/4 = 1/2. So 2 × 3 1/4 = 6 1/2. Check: 3 1/4 = 13/4, and 2 × 13/4 = 26/4 = 6 1/2.
x1_s2	Set 3: Which explanation is correct? Why does 4 × 3/8 equal 12/8?	**4 groups of 3 eighths is 12 eighths**	4 × 3/8 means 3/8 + 3/8 + 3/8 + 3/8. That is 12 eighths, or 12/8. The size of the pieces (eighths) does not change. 12/32 is a different amount.
x2_s2	Set 3: Find the mistake: Mateo says 1/2 × 4 1/2 = 2 1/2. What did he do wrong?	**He kept the 1/2 instead of multiplying it too. 1/2 × 4 1/2 = 9/4 = 2 1/4.**	Mateo found 1/2 × 4 = 2 but did not take half of the 1/2. Change 4 1/2 to 9/2. Then 1/2 × 9/2 = 9/4 = 2 1/4.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix the sentence: killa said, half of the terrace is planted with potatoes	**Killa said, "Half of the terrace is planted with potatoes."**	Capitalize the name Killa and the first word of the quote, put quotation marks around her exact words, and put the period inside the closing quotation mark.
s2_s0	Set 1: Fix the sentence: the andes mountains run through peru.	**The Andes Mountains run through Peru. OR The Andes mountains run through Peru.**	Capitalize the first word of the sentence and the names of places: the Andes and Peru.
s3_s0	Set 1: Fix the sentence: The alpaca shook it's wool after the rain.	**The alpaca shook its wool after the rain.**	Its (no apostrophe) shows ownership: the wool belongs to the alpaca. It's always means it is.
s4_s0	Set 1: Fix the sentence: the weavers in the village uses alpaca wool	**The weavers in the village use alpaca wool. OR The weaver in the village uses alpaca wool.**	Capitalize the first word. The subject is the weavers (more than one), so use the verb use, not uses. End with a period.
s1_s1	Set 2: Fix the sentence: inti asked, how many kilograms of quinoa did we harvest	**Inti asked, "How many kilograms of quinoa did we harvest?"**	Capitalize the name Inti and the first word of the quote, put quotation marks around his exact words, and put the question mark inside the closing quotation mark.
s2_s1	Set 2: Fix the sentence: Killa sells potatoes quinoa and wool at the market.	**Killa sells potatoes, quinoa, and wool at the market. OR Killa sells potatoes, quinoa and wool at the market.**	Use commas to separate the items in a series: potatoes, quinoa, and wool.
s3_s1	Set 2: Fix the sentence: many families in the village speak quechua and spanish.	**Many families in the village speak Quechua and Spanish.**	Capitalize the first word of the sentence and the names of languages: Quechua and Spanish.
s4_s1	Set 2: Fix the sentence: Each of the bags weigh 2 1/2 kilograms.	**Each of the bags weighs 2 1/2 kilograms.**	The subject is each (one bag at a time), which is singular, so use weighs, not weigh.
s1_s2	Set 3: Fix the sentence: grandmother rosa said, three fourths of the yarn is red	**Grandmother Rosa said, "Three fourths of the yarn is red."**	Capitalize Grandmother Rosa (a title used with a name) and the first word of the quote, put quotation marks around her exact words, and put the period inside the closing quotation mark.
s2_s2	Set 3: Fix the sentence: Its market day in Pisac, so the plaza is busy.	**It's market day in Pisac, so the plaza is busy.**	It's means it is: it is market day. Use the apostrophe.
s3_s2	Set 3: Fix the sentence: The market is open on sunday in the town of pisac.	**The market is open on Sunday in the town of Pisac.**	Capitalize days of the week (Sunday) and the names of towns (Pisac).
s4_s2	Set 3: Fix the sentence: The potatoes, the quinoa, and the wool is ready for market.	**The potatoes, the quinoa, and the wool are ready for market.**	The subject is three things (the potatoes, the quinoa, and the wool), so use the plural verb are, not is.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH, 10 TOTAL)

ID	Question	Answer	Why (explain to student)
match1_s0	Set 1: 3 × 1/4	**3/4**	3 × 1 = 3, so 3 × 1/4 = 3/4.
match2_s0	Set 1: 1/2 × 1/3	**1/6**	1 × 1 = 1 and 2 × 3 = 6, so 1/6.
match3_s0	Set 1: 2/3 × 3/5	**2/5**	2 × 3 = 6 and 3 × 5 = 15, so 6/15, which simplifies to 2/5 (GCF 3).
match4_s0	Set 1: 2 × 1 1/2	**3**	1 1/2 = 3/2, and 2 × 3/2 = 6/2 = 3.
match5_s0	Set 1: 7/2 as a mixed number	**3 1/2**	7 ÷ 2 = 3 R 1, so 3 1/2.
match1_s1	Set 2: 4 × 1/5	**4/5**	4 × 1 = 4, so 4 × 1/5 = 4/5.
match2_s1	Set 2: 1/4 × 1/2	**1/8**	1 × 1 = 1 and 4 × 2 = 8, so 1/8.
match3_s1	Set 2: 3/4 × 2/3	**1/2**	3 × 2 = 6 and 4 × 3 = 12, so 6/12, which simplifies to 1/2 (GCF 6).
match4_s1	Set 2: 3 × 1 1/3	**4**	1 1/3 = 4/3, and 3 × 4/3 = 12/3 = 4.
match5_s1	Set 2: 9/4 as a mixed number	**2 1/4**	9 ÷ 4 = 2 R 1, so 2 1/4.
match1_s2	Set 3: 5 × 1/6	**5/6**	5 × 1 = 5, so 5 × 1/6 = 5/6.
match2_s2	Set 3: 1/3 × 1/3	**1/9**	1 × 1 = 1 and 3 × 3 = 9, so 1/9.
match3_s2	Set 3: 4/5 × 5/8	**1/2**	4 × 5 = 20 and 5 × 8 = 40, so 20/40, which simplifies to 1/2 (GCF 20).
match4_s2	Set 3: 2 × 2 1/2	**5**	2 1/2 = 5/2, and 2 × 5/2 = 10/2 = 5.
match5_s2	Set 3: 11/3 as a mixed number	**3 2/3**	11 ÷ 3 = 3 R 2, so 3 2/3.

BOSS BATTLE: SUNDAY MARKET IN PISAC (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: It is market day in Pisac! Killa's family sells potatoes in bags that weigh 2 1/2 kilograms each. A restaurant cook buys 4 bags. Back in her kitchen, she uses 2/5 of all the potatoes she bought to make papa a la huancaína, potatoes in a creamy, spicy cheese sauce. Step 1: how many kilograms of potatoes does the cook buy? Kilograms bought:	**10 kilograms (also accept with or without the unit kg / kilogram(s))**	4 × 2 1/2 = 4 × 5/2 = 20/2 = 10 kilograms. (Or: 4 × 2 = 8 and 4 × 1/2 = 2, so 8 + 2 = 10.)
e2_s0	Set 1: It is market day in Pisac! Killa's family sells potatoes in bags that weigh 2 1/2 kilograms each. A restaurant cook buys 4 bags. Back in her kitchen, she uses 2/5 of all the potatoes she bought to make papa a la huancaína, potatoes in a creamy, spicy cheese sauce. Step 2: how many kilograms of potatoes does she use for the dish?	**4 kilograms**	She uses 2/5 of the 10 kilograms: 2/5 × 10 = 20/5 = 4 kilograms. (6 kilograms is the amount left over, and 1 kilogram is 2/5 of just one bag.)

ID	Question	Answer	Why (explain to student)
e1_s1	Set 2: It is market day in Pisac! Killa's family sells potatoes in bags that weigh 1 1/2 kilograms each. A restaurant cook buys 6 bags. Back in her kitchen, she uses 2/3 of all the potatoes she bought to make papa a la huancaína, potatoes in a creamy, spicy cheese sauce. Step 1: how many kilograms of potatoes does the cook buy? Kilograms bought:	**9 kilograms (also accept with or without the unit kg / kilogram(s))**	6 × 1 1/2 = 6 × 3/2 = 18/2 = 9 kilograms. (Or: 6 × 1 = 6 and 6 × 1/2 = 3, so 6 + 3 = 9.)
e2_s1	Set 2: It is market day in Pisac! Killa's family sells potatoes in bags that weigh 1 1/2 kilograms each. A restaurant cook buys 6 bags. Back in her kitchen, she uses 2/3 of all the potatoes she bought to make papa a la huancaína, potatoes in a creamy, spicy cheese sauce. Step 2: how many kilograms of potatoes does she use for the dish?	**6 kilograms**	She uses 2/3 of the 9 kilograms: 2/3 × 9 = 18/3 = 6 kilograms. (3 kilograms is the amount left over, and 1 kilogram is 2/3 of just one bag.)
e1_s2	Set 3: It is market day in Pisac! Killa's family sells potatoes in bags that weigh 3 1/4 kilograms each. A restaurant cook buys 4 bags. Back in her kitchen, she uses 3/4 of all the potatoes she bought to make papa a la huancaína, potatoes in a creamy, spicy cheese sauce. Step 1: how many kilograms of potatoes does the cook buy? Kilograms bought:	**13 kilograms (also accept with or without the unit kg / kilogram(s))**	4 × 3 1/4 = 4 × 13/4 = 52/4 = 13 kilograms. (Or: 4 × 3 = 12 and 4 × 1/4 = 1, so 12 + 1 = 13.)
e2_s2	Set 3: It is market day in Pisac! Killa's family sells potatoes in bags that weigh 3 1/4 kilograms each. A restaurant cook buys 4 bags. Back in her kitchen, she uses 3/4 of all the potatoes she bought to make papa a la huancaína, potatoes in a creamy, spicy cheese sauce. Step 2: how many kilograms of potatoes does she use for the dish?	**9 3/4 kilograms**	She uses 3/4 of the 13 kilograms: 3/4 × 13 = 39/4 = 9 3/4 kilograms. (3 1/4 kilograms is the amount left over.)

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	3 baskets, each 1/4 full	1/4 + 1/4 + 1/4 = 3/4 and 3 × 1/4 = 3/4. She has 3/4 of a basket of potatoes in all.
2	Which is the same as 4 × 1/5?	4/5 (4 groups of 1 fifth is 4 fifths).
3	1/2 of 8 and 1/4 of 8	1/2 of 8 = 4 alpacas; 1/4 of 8 = 2 alpacas.
4	Area model: 1 of 4 columns, 1 of 2 rows	1/8 of the rectangle is shaded both colors (1 of 8 equal parts). This is 1/2 × 1/4 = 1/8.
5	Simplify 8/12; write 9/4 as a mixed number	8/12 = 2/3 (GCF 4); 9/4 = 2 1/4.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	6 ponchos × 5/8 kilogram	6 × 5/8 = 30/8 = 15/4 = 3 3/4 kilograms.
2	Area model for 2/3 × 3/5	Rectangle with 5 columns (3 shaded) and 3 rows (2 shaded); 6 of 15 parts overlap. 2/3 × 3/5 = 6/15 = 2/5 (GCF 3).
3	Which product is less than 4/5?	1/2 × 4/5 (= 2/5). Multiplying by a number less than 1 gives a smaller product; 1 × 4/5 equals 4/5, and 2 and 1 1/2 make it bigger.
4	3 bags × 2 1/3 kilograms, half used for soup	3 × 2 1/3 = 3 × 7/3 = 21/3 = 7 kilograms bought. 1/2 × 7 = 7/2 = 3 1/2 kilograms in the soup.
5	Writing: is the product always bigger?	He is not right. Model: "Inti is wrong, because multiplying by a fraction less than 1 gives a smaller answer. For example, 6 × 1/2 = 3, and 3 is less than 6. 1/2 × 1/2 = 1/4, which is less than 1/2." Look for complete sentences, a correct example, and the idea that multiplying by a fraction less than 1 takes only part of a number.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Purple potatoes ready to harvest	1/4 of the terrace. 1/2 × 2/3 × 3/4 = 6/24 = 1/4. (Purple: 2/3 × 3/4 = 6/12 = 1/2 of the terrace; half of that is 1/4.)